

Bank Marketing Data Analysis

This notebook presents an analysis of the `bankmarketing.csv` dataset, which includes data related to a bank's marketing campaigns. The main goal is to understand customer behavior and predict whether a client will subscribe to a term deposit.

```
import pandas as pd

# Load the dataset
df = pd.read_csv('bankmarketing.csv')

# Display the first few rows
df.head()
```

	age	job	marital	education	default	housing	loan	contact	month	day_of_week	...	campaign	pdays	previous	poutcome
0	56	housemaid	married	basic.4y	no	no	no	telephone	may	mon	...	1	999	0	none
1	57	services	married	high.school	unknown	no	no	telephone	may	mon	...	1	999	0	none
2	37	services	married	high.school	no	yes	no	telephone	may	mon	...	1	999	0	none
3	40	admin.	married	basic.6y	no	no	no	telephone	may	mon	...	1	999	0	none
4	56	services	married	high.school	no	no	yes	telephone	may	mon	...	1	999	0	none

5 rows × 21 columns

```
# Check for missing values and data types
print(df.info())
print(df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   age                   41188 non-null  int64
1   job                   41188 non-null  object
2   marital               41188 non-null  object
3   education             41188 non-null  object
4   default               41188 non-null  object
5   housing               41188 non-null  object
6   loan                  41188 non-null  object
7   contact               41188 non-null  object
8   month                 41188 non-null  object
9   day_of_week           41188 non-null  object
10  duration              41188 non-null  int64
11  campaign              41188 non-null  int64
12  pdays                41188 non-null  int64
13  previous              41188 non-null  int64
14  poutcome              41188 non-null  object
15  emp.var.rate          41188 non-null  float64
16  cons.price.idx         41188 non-null  float64
17  cons.conf.idx         41188 non-null  float64
18  euribor3m             41188 non-null  float64
19  nr.employed           41188 non-null  float64
20  y                     41188 non-null  object
dtypes: float64(5), int64(5), object(11)
memory usage: 6.6+ MB
None
age                0
job                0
marital            0
education          0
default            0
housing            0
loan               0
contact            0
month              0
day_of_week        0
duration           0
```

```
campaign      0
pdays        0
previous      0
poutcome      0
emp.var.rate  0
cons.price.idx 0
cons.conf.idx 0
euribor3m     0
nr.employed   0
y             0
dtype: int64
```

```
# Summary statistics

df.describe()
```

	age	duration	campaign	pdays	previous	emp.var.rate	cons.price.idx	cons.conf.idx	euribor3
count	41188.00000	41188.000000	41188.000000	41188.000000	41188.000000	41188.000000	41188.000000	41188.000000	41188.00000
mean	40.02406	258.285010	2.567593	962.475454	0.172963	0.081886	93.575664	-40.502600	3.62129
std	10.42125	259.279249	2.770014	186.910907	0.494901	1.570960	0.578840	4.628198	1.73444
min	17.00000	0.000000	1.000000	0.000000	0.000000	-3.400000	92.201000	-50.800000	0.63400
25%	32.00000	102.000000	1.000000	999.000000	0.000000	-1.800000	93.075000	-42.700000	1.34400
50%	38.00000	180.000000	2.000000	999.000000	0.000000	1.100000	93.749000	-41.800000	4.85700
75%	47.00000	319.000000	3.000000	999.000000	0.000000	1.400000	93.994000	-36.400000	4.96100
max	98.00000	4918.000000	56.000000	999.000000	7.000000	1.400000	94.767000	-26.900000	5.04500

```
import seaborn as sns
import matplotlib.pyplot as plt

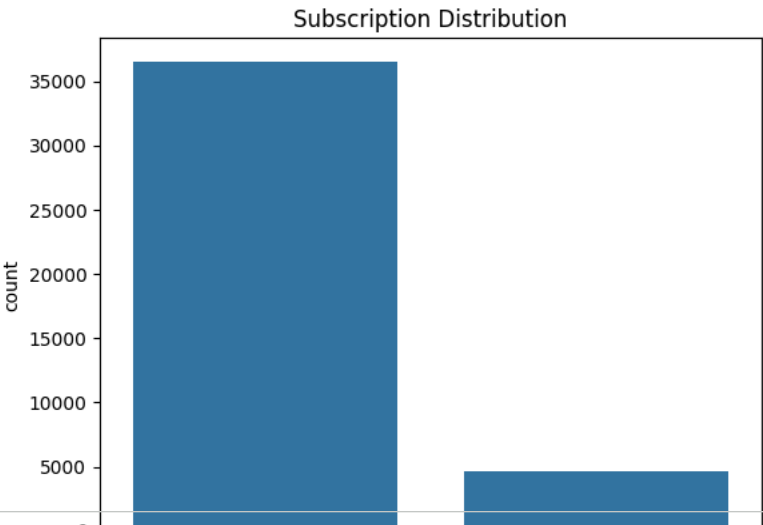
# Plot the distribution of the target variable

sns.countplot(x='y', data=df)

plt.title('Subscription Distribution')

plt.show()

# Plot the distribution of the target variable
```



```
# Correlation matrix for numerical features

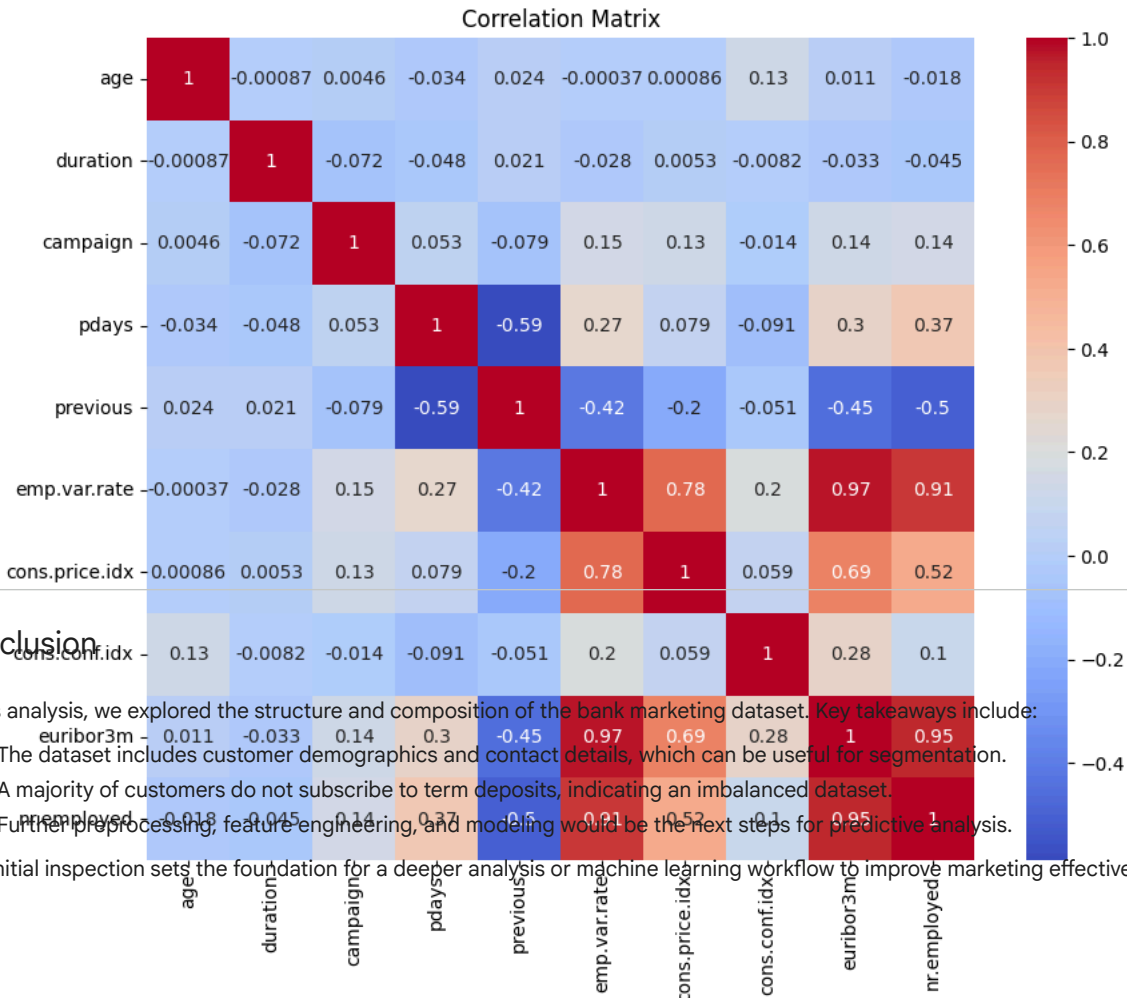
plt.figure(figsize=(10,8))

sns.heatmap(
```

```
df.corr(numeric_only=True),
annot=True,
cmap='coolwarm'
)

plt.title('Correlation Matrix')

plt.show()
```



Conclusion

In this analysis, we explored the structure and composition of the bank marketing dataset. Key takeaways include:

- The dataset includes customer demographics and contact details, which can be useful for segmentation.
- A majority of customers do not subscribe to term deposits, indicating an imbalanced dataset.
- Further preprocessing, feature engineering, and modeling would be the next steps for predictive analysis.

This initial inspection sets the foundation for a deeper analysis or machine learning workflow to improve marketing effectiveness.